

Cogal Jacques Tracy Pasaol

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EDUCATION

Nanyang Technological University, Singapore

Aug 2020 - Jun 2024

Double Honours Degree Programme (Distinctions)

Bachelor of Engineering (Computer Science) and

Bachelor of Business (with specialization in Business Analytics)

Temasek Polytechnic

Apr 2015 - Apr 2018

Diploma in Accounting and Finance (with Merit)

WORK EXPERIENCE

UBS

Singapore

Software Engineer | AI Engineer

Feb 2025 - Present

Stack: Kotlin, Python, Kafka, Langchain, Azure

Wealth Management Core Platform Team and Agentic Products Team

- Led development of a greenfield Kafka-based transaction system processing hundreds of thousands of daily transactions for Credit Suisse migration
- Reduced snapshot processing time from 5 hours to 20 minutes (~15x) using Kotlin coroutines
- Built a multi-agent LLM pipeline using LangChain and Azure OpenAI for document analysis and cross-source validation in KYC workflows, reducing processing time from months to days
- Led team to win UBS Global Hackathon 2025 by developing an agentic system for automated regulatory event processing

Shopee

Singapore

Software Engineer

Jun 2024 - Feb 2025

Stack: Golang, Protobuf, MySQL, Kafka

Backend Engineer under marketplace core team

- Developed backend services in Golang for Shopee's order fulfilment system, processing millions of transactions per hour across the full order lifecycle
- Designed stateful order lifecycle processing with idempotency, retries, and timeout handling to ensure consistency across distributed services
- Built event-driven pipelines using Kafka (CDC) to propagate database changes and enable asynchronous processing across microservices

CPF Board

Singapore

Software Engineer Intern

Dec 2023 - May 2024

Stack: Python, JavaScript (Next.js), Azure

Involved in projects relating to large language models and data engineering

- Developed full-stack applications using Next.js to improve internal workflows and public-facing services
- Built an LLM-powered search system using vector embeddings to improve internal knowledge base accessibility and navigation
- Improved search relevance (MRR) by ~10% for a customer-facing search system

JP Morgan Chase & Co.

Singapore

Software Engineer Intern

Jun 2023 - Aug 2023

Stack: Python, Java, JavaScript (React.js), Amazon Web Services

Delivered projects in artificial intelligence and proxy gateway solutions for asset wealth management teams

- Engineered data pipelines to feed large-scale datasets into AI models for risk assessment
- Developed an SDK to proxy microservices via a caching layer, reducing gateway calls by over 90%
- Led team to win JP Morgan Global Hackathon by building an anti-money laundering solution

AWARDS AND HONOURS

Champion of UBS Global Hackathon 2025 Led team in building AI RAG solution for regulatory event triaging and impact assessment	2025
Champion of GIC Code to Impact 2023 Led team to win competition with AI enabled full stack solution	2023
Dean's Honours List AY2022/2023 (BUS) Top 5% of student cohort based on annual academic performance	2023
Champion of J.P Morgan Global Hackathon Singapore 2023 Led team to win competition with a solution that leverages transformer models	2023
Champion of J.P Morgan Code for Good 2022 Developed full-stack solution to address social enterprise problem	2022

PROJECTS

Tutor LLM (Retrieval Augmented Generation (RAG) Community Knowledgebase)

Stack: Golang, Python, React Typescript, MySQL, Kafka, Weaviate, Docker, gRPC, Protobuf and Redis

GitHub: [Link](#)

Developed a full-stack webapp with a distributed backend system for community members to share RAG knowledge base partitions amongst each other

- Decoupled heavy processing tasks such as parsing and embedding documents with Kafka
- Implemented Weaviate as a vector database for efficient semantic search and pre-filtering
- Reduced latency by 20% with gRPC and protobuf for lightweight server communication
- Achieved dynamic learning material creation with structured AI outputs

CN LLM (Learn HSK 1-6 Chinese with AI generated flash cards and evaluation)

Stack: Typescript React, Python, MySQL

GitHub: [Link](#)

Learning Chinese with dynamically generated flash cards and space repetition scheduler

- Implemented space repetition scheduler with FSRS to maximise retention rate and optimise learning

Portfolio Chatbot (Retrieval Augmented Generation (RAG) AI Chatbot)

Stack: React, Python Flask, LangChain, OpenAI, AWS EC2, Docker and Redis Vector DB

Site: jacquescogal.github.io

Built a personal portfolio with an AI chatbot that delivers context rich answers about me using RAG

- Cut vector database retrieval time from seconds to milliseconds by migrating to Redis vector store

SKILLS

Languages: Go, Python, Kotlin, Java, TypeScript

Backend & Distributed Systems: Kafka, Microservices, Event-Driven Architecture, gRPC, REST, Protobuf, CDC

Databases: MySQL, PostgreSQL, DynamoDB, Cosmos DB, Redis, Weaviate (Vector DB)

AI: LLMs, RAG, LangChain, Vector Embeddings, Multi-Agent Systems

System Design: Scalability, Fault Tolerance, Idempotency, Distributed Transactions, Async Processing

CERTIFICATES

Azure AI Engineer Associate (Credential ID: 364C9693233635FA)	2026
Azure Developer Associate (Credential ID: EA3E3DE0F075D818)	2026
Azure Administrator Associate (Credential ID: 60E23EE7F800BE17)	2026